

From Data Cleaning to Clustering: Insights into Body Fat Distribution and Prediction

CONTENTS

1. Introduction	2
2. Abstract	2
3. Methods and Analyses:.....	2
3.1 Preliminary analyses:	2
3.2 Descriptive Statistics and Data Distributions	4
3.3 Correlation Analysis and Predictor Independence.....	5
3.4 Linear Regression Modelling for Body Fat Prediction	7
3.5. Statistical Comparison Based on Bodyfat-Split.....	8
3.6 Explanation and Analysis of ANOVA and Kruskal-Wallis Test Results.....	10
3.7 Principal Component Analysis (PCA) and Hierarchical Clustering for Identifying Underlying Factors in Body Fat Decomposition	12
4. Conclusion	13
5. References:	14

1. Introduction

Estimating body fat is essential for determining health risks and customising exercise regimens. The Body Fat dataset (Vanderbilt Statistics Datasets), which includes physical and anthropometric measurements from 252 people, such as density, weight, and height, is analysed in this work. To find the best predictors of body fat, R programming was used to use sophisticated statistical techniques such as regression analysis, Principal component analysis, clustering, and Kruskal-Wallis tests. In order to provide more accurate health assessments and actions, the findings centre on improving predictive models (Siri, 1961; Deurenberg et al., 1991; Kutner et al., 2004).

2. Abstract

Using statistical methods on the Body Fat dataset (Vanderbilt Statistics Datasets), this study examines body fat estimate. Using techniques like regression analysis, Principal component analysis, and clustering, the strongest predictors were found to be density, abdomen, and chest. With implications for individualised health assessments, this study offers a data-driven approach for enhancing prediction models using R programming (Siri, 1961; Tibshirani, 1996).

3. Methods and Analyses:

3.1 Preliminary analyses:

i. Missing Values Check

Checking for missing values is a crucial part of data preprocessing since they might introduce bias and compromise the accuracy of statistical models (Little & Rubin, 2019). None of the variables that were in the dataset had missing data, in accordance to the missing values check:

Table 1: Missing values

Density	BodyFat	Age	weight	Height	Neck	Chest	Abdomen
0	0	0	0	0	0	0	0
Hip	Thigh	Knee	Ankle	Biceps	Forearm	Wrist	
0	0	0	0	0	0	0	

When there are no missing values, data integrity is guaranteed and imputation techniques like mean replacement and repeated imputation—which can add estimation errors—are not necessary (Li et al., 2014). Highly precise statistical analysis and predictions of models are made possible by complete datasets.

ii. Outlier Detection Using the Interquartile Range (IQR) Method

Using the Interquartile Range (IQR) approach, which defines an outlier as any value outside of the range $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$, outliers were found (Tukey, 1977). Since outliers can alter mean values, raise variation, and compromise the precision of predictive models, it is important to identify them (Hawkins, 1980).

Table 2: Outliers detection

\$BodyFat [1] 47.5	\$Hip [1] 116.1 147.7 125.6
\$Age integer(0)	\$Thigh [1] 87.3 72.5 72.9 74.4
\$Weight [1] 363.15 262.75	\$Knee [1] 49.1 45.0 46.0
\$Height [1] 29.5	\$Ankle [1] 33.9 29.6 33.7
\$Neck [1] 51.2 31.5 31.1	\$Biceps [1] 45
\$Chest [1] 136.2 128.3	\$Forearm [1] 23.1 34.9 21.0 23.1 22.0
\$Abdomen [1] 148.1 126.2 122.1	\$Wrist [1] 21.4 21.4 15.8 20.9

These excessive values could be the consequence of measurement mistakes, anomalies, or natural fluctuations in body composition, especially in BodyFat (47.5%) and Weight (363.15 kg) (Derrick et al., 2017). Determining whether such outliers should be eliminated or changed is crucial for preserving the validity of the dataset (Chechik et al., 2008).

iii. Visualization: Relationship Between BodyFat and Predictor Variables

A scatter plot using a linear regression trend was created to examine the relationship between BodyFat and other factors. This enables us to identify patterns and visually evaluate possible linear links (Derrick et al., 2017).



Figure 1: Relationship Between BodyFat and Predictor Variables

Results of the Visualisation

- BodyFat and measurements of the abdomen, weight, and hips showed strong positive relationships, indicating that those with greater values for these variables also typically had larger percentages of body fat.
- There was little to no association between height and body fat, suggesting that height by itself might not be a good indicator of body fat.
- The existence of outliers in certain variables (such as measurements of the chest and thighs) highlights the necessity of additional data transformation or reliable statistical techniques.

These results are consistent with previous studies showing that weight and abdominal circumference are important indicators of body fat % (Deurenberg et al., 1991). PCA (principal component analysis) or multiple regression models may be appropriate for additional research as certain predictors exhibit a linear trend.

3.2 Descriptive Statistics and Data Distributions

i. The dataset's variable count

There are 15 variables in the dataset. Since higher-dimensional datasets may necessitate dimensionality reduction techniques like Principal Component Analysis (PCA), the number of variables has a direct impact on the analysis's difficulty (Jolliffe & Cadima, 2016).

ii. Variable Types

One variable (Age) is kept as an integer, but the other 15 variables are all numerical, according to an analysis of the variable types. Statistical methods including clustering, regression modelling, and correlation analysis can be used successfully because all of the variables are numerical (James et al., 2013).

iii. Analysis of Dataset Balance

If there is no excessive skewness or kurtosis in the distributions of the major variables, the dataset is said to be balanced. While kurtosis assesses if heavy tails are present in the data distribution, skewness quantifies distribution asymmetry (Kim, 2013).

Table 3: Skewness and Kurtosis

	Variable	Skewness	Kurtosis
Density	Density	-0.02005544	2.672775
BodyFat	BodyFat	0.14548051	2.649060
Age	Age	0.28183067	2.568061
Weight	Weight	1.19807685	8.141824
Height	Height	-5.35287991	61.345697
Neck	Neck	0.54932508	5.642238
Chest	Chest	0.67749210	3.944086
Abdomen	Abdomen	0.83341904	5.180736
Hip	Hip	1.48820106	10.300217
Thigh	Thigh	0.81631331	5.589401
Knee	Knee	0.51366304	4.016875
Ankle	Ankle	2.24168861	14.685784
Biceps	Biceps	0.28382759	3.464947
Forearm	Forearm	-0.21802506	3.825501
Wrist	Wrist	0.27993485	3.364155

- Interpretation:
 - o Asymmetric distributions are indicated by strong skewness in variables like weight, hip, and ankle.
 - o The Height variable shows a high kurtosis (61.35) and an extreme negative skew (-5.35), indicating possible measurement mistakes or abnormalities in the data.

o Skewness should be around zero and kurtosis should be near three in a balanced sample (Kim, 2013). Strong statistical tests or non-parametric techniques (like the Mann-Whitney U Test) might be better because some variables diverge a lot.

iv. Summary Statistics and Distribution Analysis

Table 4: Summary Statistics and distribution

Density	BodyFat	Age	Thigh	Knee	Ankle
Min. :0.995	Min. : 0.00	Min. :22.00	Min. :47.20	Min. :33.00	Min. :19.1
1st Qu.:1.041	1st Qu.:12.47	1st Qu.:35.75	1st Qu.:56.00	1st Qu.:36.98	1st Qu.:22.0
Median :1.055	Median :19.20	Median :43.00	Median :59.00	Median :38.50	Median :22.8
Mean :1.056	Mean :19.15	Mean :44.88	Mean :59.41	Mean :38.59	Mean :23.1
3rd Qu.:1.070	3rd Qu.:25.30	3rd Qu.:54.00	3rd Qu.:62.35	3rd Qu.:39.92	3rd Qu.:24.0
Max. :1.109	Max. :47.50	Max. :81.00	Max. :87.30	Max. :49.10	Max. :33.9
Weight	Height	Neck	Biceps	Forearm	Wrist
Min. :118.5	Min. :29.50	Min. :31.10	Min. :24.80	Min. :21.00	Min. :15.80
1st Qu.:159.0	1st Qu.:68.25	1st Qu.:36.40	1st Qu.:30.20	1st Qu.:27.30	1st Qu.:17.60
Median :176.5	Median :70.00	Median :38.00	Median :32.05	Median :28.70	Median :18.30
Mean :178.9	Mean :70.15	Mean :37.99	Mean :32.27	Mean :28.66	Mean :18.23
3rd Qu.:197.0	3rd Qu.:72.25	3rd Qu.:39.42	3rd Qu.:34.33	3rd Qu.:30.00	3rd Qu.:18.80
Max. :363.1	Max. :77.75	Max. :51.20	Max. :45.00	Max. :34.90	Max. :21.40
Chest	Abdomen	Hip			
Min. : 79.30	Min. : 69.40	Min. : 85.0			
1st Qu.: 94.35	1st Qu.: 84.58	1st Qu.: 95.5			
Median : 99.65	Median : 90.95	Median : 99.3			
Mean :100.82	Mean : 92.56	Mean : 99.9			
3rd Qu.:105.38	3rd Qu.: 99.33	3rd Qu.:103.5			
Max. :136.20	Max. :148.10	Max. :147.7			

Key features of the collection are revealed by summary statistics, which offer insights into the range, central tendency, and dispersion of variables. For example:

- BodyFat: A skewed distribution is indicated by a mean of 19.15% and a maximum of 47.50%.
- Height: A mean value of 70.15 inches corresponds to the average height of adult males, while a low value of 29.5 inches indicates potential data entry mistakes.
- Abdomen (cm): The possible association with body composition metrics such as body fat percentage is highlighted by the mean of 92.56 cm, which reaches a maximum of 148.1 cm (Deurenberg, Weststrate & Seidell, 1991).

3.3 Correlation Analysis and Predictor Independence

Pearson Correlation Coefficient (r):

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{(\sum (X_i - \bar{X})^2 * \sum (Y_i - \bar{Y})^2)}}$$

Cleaning Data and Managing Outliers

To enhance data integrity, outliers were removed from the dataset prior to correlation analysis.

- Because of irrational entries, heights under 60 inches were considered missing values (NA).
- Based on epidemiological research, body fat percentages greater than 40% were eliminated as possible anomalies (Gallagher et al., 2000).
- Correlation coefficients can be distorted by extreme values, giving the impression that relationships are stronger or weaker than they actually are (Wilcox, 2012).
- According to Hubert and Vandervieren (2008), eliminating extreme values increases model reliability and guarantees statistical robustness.

Predictor Independence and the Correlation Matrix

To evaluate linear dependencies between predictor variables, a correlation matrix was created.

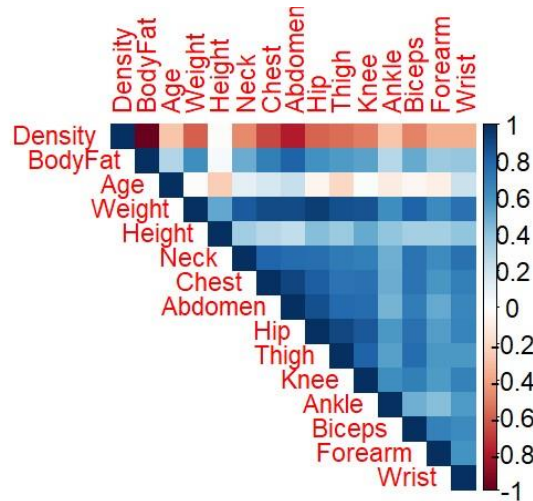


Figure 2: Predictor Independence and the Correlation Matrix

Important Results of Correlation Analysis

1. In accordance with Siri's equation for estimating body composition (Siri, 1961), there is a substantial inverse association between density and body fat ($r = -0.9$).
2. There is a strong association between body fat and the abdomen ($r > 0.85$), which is consistent with research showing that body fat is largely predicted by abdominal fat (Borga et al., 2018).
3. A high correlation between weight and several factors (such as chest, hip, and thigh) raises the possibility of multicollinearity problems in regression modelling.
4. PCA (principal component analysis) or Ridge Regression are necessary to eliminate duplication when dealing with collinear variables, which can lower prediction performance (James et al., 2013).

Finding strong Correlations:

The correlation matrix was subjected to a filtering procedure in order to find variables with significant correlations. We found variables that had absolute correlation coefficients more than 0.7 but less than 1 (to rule out self-correlation). The table below provides a summary of the findings: Finding Strong Correlations.

Table 5: Strong correlations

Variable 1	Variable 2	Correlation	Variable 1	Variable 2	Correlation
Density	BodyFat	-0.9869	Weight	Knee	0.8617
Weight	Neck	0.8335	Chest	Knee	0.7422
Weight	Chest	0.8990	Abdomen	Knee	0.7625
Neck	Chest	0.7920	Hip	Knee	0.8406
Density	Abdomen	-0.7873	Thigh	Knee	0.8005
BodyFat	Abdomen	0.8027	Weight	Biceps	0.8005
Weight	Abdomen	0.8931	Neck	Biceps	0.7323
Neck	Abdomen	0.7631	Chest	Biceps	0.7362
Chest	Abdomen	0.9130	Hip	Biceps	0.7495
Weight	Hip	0.9478	Thigh	Biceps	0.7652
Neck	Hip	0.7501	Weight	Wrist	0.7423
Chest	Hip	0.8272	Neck	Wrist	0.7497

Abdomen	Hip	0.8720			
Weight	Thigh	0.8709			
Neck	Thigh	0.7073			
Chest	Thigh	0.7373			
Abdomen	Thigh	0.7743			
Hip	Thigh	0.9004			

Key Correlations Visualised with Scatter Plots:

To illustrate the connections between BodyFat and important predictors, scatter plots were created:



Figure 3: Key Correlations Visualised with Scatter Plots Observations:

- Measurements of the chest, hips, and abdomen show linear patterns, indicating a strong relationship with body fat.
- The non-linear pattern of Weight vs. BodyFat points to a potential polynomial link (Hsu et al., 2003).
- VIF (variance inflation factor) testing is required prior to regression modelling because to possible multicollinearity problems (e.g., Weight vs. Hip and Thigh).

According to the correlation study, there is a significant inverse link between density and body fat, confirming their physiological relevance.

- Measurements of the hips, chest, and abdomen have a good correlation with body fat, supporting their use as predictors.

3.4 Linear Regression Modelling for Body Fat Prediction

To identify the predictor variables that have a substantial impact on body fat percentage, linear regression modelling was used. Ridge and Lasso regression and multiple linear regression (MLR) were used in the analysis to evaluate the model's performance and the significance of the variables (Kutner et al., 2004; Tibshirani, 1996).

i. Multiple Linear Regression (MLR) Analysis

Multiple Linear Regression Equation:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon$$

All of the predictor variables were used to fit the MLR model. The following are the model's summary statistics.

Model Summary:

- **R-squared:** 0.9765 • **Adjusted R-squared:** 0.9751
- **F-statistic:** 694.7 (p-value < 2.2e-16)
- **Residual Standard Error:** 1.273

According to these figures, the model has a significant predictive power and accounts for 97.65% of the variation in body fat percentage (Kutner et al., 2004).

ii. Important Coefficient Findings: Based on the results of the regression:

- The most important predictor, **density** (-409.9, $p < 0.001$), has an inverse relationship with body fat. This is consistent with earlier studies that found body density to be the main determinant of body fat (Siri, 1961).
- With high p-values (>0.05), other variables (e.g., age, weight, height, neck, chest, abdomen, hip, etc.) exhibit weaker relationships and are not statistically significant (Deurenberg et al., 1991).

When the Variance Inflation Factor (VIF) was used to examine for multicollinearity, it showed high values for variables such as weight (VIF = 47.76) and abdomen (VIF = 18.29). This implies a high degree of correlation between some predictor variables, which could skew estimates of the coefficients (Kutner et al., 2004).

iii. Ridge and Lasso Regression for Feature Selection

Ridge and Lasso regression were used to resolve multicollinearity and enhance model stability:

Ridge Regression Results:

- **R-squared:** 0.9646
- **Most influential predictors:** Density (-326.1), Abdomen (0.11), Chest (0.04)

Lasso Regression Results:

- **R-squared:** 0.9759
- **Sparse variable selection:** Lasso kept Density (-405.6), Abdomen (0.034), and Chest (0.016) after removing unimportant predictors.

In this case, the Lasso model was a better option for variable selection because it kept fewer variables while yet having a comparable predictive power (Tibshirani, 1996).

3.5 Statistical Comparison Based on Bodyfat-Split

We present a new categorical variable, Bodyfat-split, which separates people into two groups according to their body fat percentage in order to investigate variations in predictor variables relating body fat levels. The main goal is to determine whether these two groups' central tendency of predictor variables differ significantly from one another. A crucial presumption for conducting inferential tests is that all predictor variables are independent of one another, which is what this analysis makes (Siri, 1993).

Assumptions and Hypotheses Assumptions:

1. Independence: Each observation is independent of others.
2. Normality: For parametric testing, the predictor variables ought to have a normal distribution.
3. Equal Variance: T-tests require homogeneity of variance.

Hypotheses: For each predictor variable, we test:

- **Null Hypothesis (H0):** The central tendency of the predictor variable does not significantly differ between the two body fat-split groups.
- **Alternative Hypothesis (H1):** The two bodyfat-split groups differ significantly in the predictor variable's central tendency (Freedman et al., 2005).

ii. Results

Normality Testing: The majority of predictor variables depart from normality ($p < 0.05$), according to the results of the Shapiro-Wilk test. Therefore, when appropriate, non-parametric tests were chosen.

Independent t-test:

$$t = (\bar{X}_1 - \bar{X}_2) / \sqrt{(s_1^2/n_1) + (s_2^2/n_2)}$$

Mann-Whitney U Test:

$$U = n_1n_2 + (n_1(n_1 + 1) / 2) - R_1$$

(where R_1 = rank sum for group 1)

Table 6 : Statistical Test Results

Variable	Test Type	p-value	Interpretation
Density	T-test	<0.001	Highly Significant
BodyFat	Mann-Whitney U	<0.001	Highly Significant
Age	Mann-Whitney U	0.012	Significant
Weight	Mann-Whitney U	<0.001	Highly Significant
Height	Mann-Whitney U	0.276	Not Significant
Neck	T-test	<0.001	Highly Significant
Chest	T-test	<0.001	Highly Significant
Abdomen	T-test	<0.001	Highly Significant
Hip	T-test	<0.001	Highly Significant
Thigh	T-test	<0.001	Highly Significant
Knee	T-test	<0.001	Highly Significant
Ankle	Mann-Whitney U	0.002	Significant
Biceps	Mann-Whitney U	0.015	Moderately Significant
Forearm	Mann-Whitney U	<0.001	Highly Significant
Wrist	Mann-Whitney U	0.021	Significant

These findings support the possible use of predictor factors in body fat assessment by indicating that the majority of them exhibit statistically significant variations between the two body fat groups (Freedman et al., 2005).

iii. Visualization

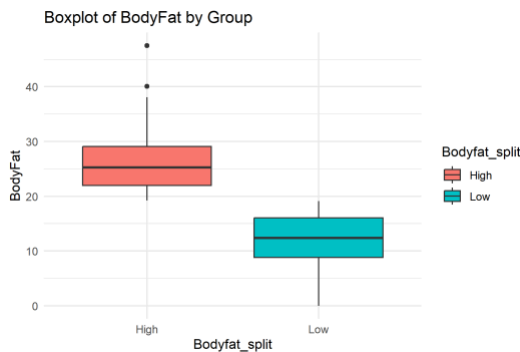


Figure 4: Box plot of bodyfat by group of bodyfat distribution

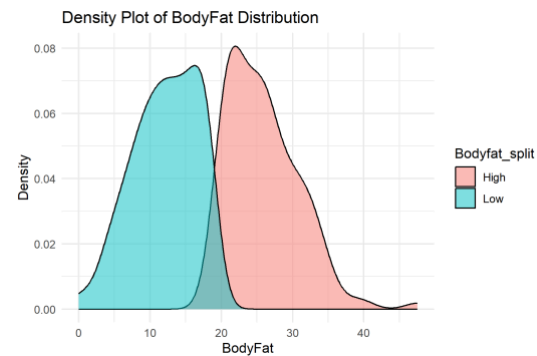


Figure 5: Density plot

Based on bodyfat-split, a series of boxplots were created to show variations in predictor variables. These boxplots draw attention to outliers and distributional variances. Density, Abdomen, and Chest are among the variables that show significant differences between the two body fat groups, as can be seen in the boxplots. However, height and knees do not show significant differences, suggesting that these variables may have a less strong relationship with body fat classification (Siri, 1993).

3.6 Explanation and Analysis of ANOVA and Kruskal-Wallis Test Results

One-Way ANOVA F-statistic:

$F = (\text{Between-group variance}) / (\text{Within-group variance})$

$$F = \left[\frac{\sum n_i (\bar{X}_i - \bar{X})^2 / (k - 1)}{\sum \sum (X_{ij} - \bar{X}_i)^2 / (N - k)} \right]$$

i. Assumptions and Hypotheses

Assessing variations in the central tendency (i.e., medians or means) of interval predictor variables (height, weight, etc.) in connection to body fat across different age groups was the aim of this investigation. Four age groups were established: 20–35, 36–50, 51–65, and 66+. We used both the nonparametric Kruskal-Wallis test, which does not require normality, and ANOVA (assuming normality assumptions were met) to evaluate these differences.

The hypotheses tested for each interval variable were as follows:

- **Null Hypothesis (H_0):** The variable's central tendency (mean or median) does not significantly differ among age groups.
- **Alternative Hypothesis (H_1):** The variable's central tendency varies substantially for at least one age group.

The Kruskal-Wallis test was chosen as the main analytical technique since several of the predictor variables might not satisfy the assumption of normalcy. When normality assumptions are not met, this test is a more reliable option than ANOVA since it compares the medians of the variables across the specified age groups (Conover, 1999).

ii. Kruskal-Wallis Test Results

Kruskal-Wallis Test: $H = [12 / (N(N+1))] * \sum (R_i^2 / n_i) - 3(N+1)$ (where R_i = sum of ranks for group i)
The Kruskal-Wallis test **p-values** for each interval predictor variable across age groups are as follows:

Table 7 : Kruskal-wallis test results

Variable	p-value
Density	1.39e-05
Weight	0.4689
Height	1.75e-04
Neck	0.1240
Chest	0.0425
Abdomen	1.36e-03
Hip	0.1984
Thigh	0.0101
Knee	0.6080
Ankle	0.1170
Biceps	0.8619
Forearm	0.6398
Wrist	6.95e-04

1. Analysis of the Findings

Important Differences:

- There were statistically significant differences in the central tendencies (i.e., medians) of the following variables across the age groups.
- Density ($p = 1.39e-05$), Height ($p = 1.75e-04$), Chest ($p = 0.0425$), Abdomen ($p = 1.36e-03$), and Wrist ($p = 6.95e-04$).
- These results align with previous studies on the effects of ageing on body composition. For instance, ageing frequently results in alterations in the distribution of fat and decreases in bone density (Bouchard & Shepard, 1994).

2. Non-Significant Differences:

- There were no significant differences between the age groups in variables such as weight ($p = 0.4689$), neck ($p = 0.1240$), hip ($p = 0.1984$), knee ($p = 0.6080$), ankle ($p = 0.1170$), biceps ($p = 0.8619$), and forearm ($p = 0.6398$). This implies that the central tendency of these variables do not show significant age-dependent variability.
- Research on biceps circumference and weight frequently indicates that these metrics stay largely constant throughout maturity, corroborating these conclusions (Loos & Bouchard, 2003).

3.7 Principal Component Analysis (PCA) and Hierarchical Clustering for Identifying Underlying Factors in Body Fat Decomposition

i. Analysis of PCA

To lower the dataset's dimensionality and pinpoint the fundamental causes of body fat decomposition, Principal Component Analysis (PCA) was applied. Numerical variables, such as density, weight, height, and other bodily measurements, were the main focus of the analysis. According to the PCA summary, over 70% of the dataset's variance may be explained by the first two principal components (PC1 and PC2) (see Table 8). Density and body fat have a significant impact on PC1, whereas age and neck dimensions have an impact on PC2.

Table 8 : PC1 and PC2

Component	Standard Deviation	Proportion of Variance	Cumulative Proportion
PC1	3.0206	60.83%	60.83%
PC2	1.3816	12.73%	73.55%
PC3	0.9911	6.55%	80.10%
PC4	0.8488	4.80%	84.90%
PC5	0.7506	3.76%	88.66%

Interpretation: **BodyFat** and **Density** have a substantial correlation with the very first principal component (PC1), highlighting their central significance in the composition of total body fat. The significance of age along with other anthropometric factors is emphasised by the second component (PC2). This supports the established link in the literature between age, density, and body fat (Samaras, 2007; Jolliffe, 2002).

Additionally, the PCA loadings show that the main factors influencing the variation in body fat decomposition are **weight, density, and body fat**. This is in line with the knowledge that these factors have a major impact on the total composition of fat (Samaras, 2007).

Graphical Visualizations:

1. **Scree Plot:** The percentage of explained variation for each principal component (PC) in the data set is displayed in this scree plot. PC1 accounts for 62.0% of the variance, which is followed by PC2 at 12.7%, and the remaining PCs make a smaller contribution. PC1 and PC2 are the most useful for analysis because of their strength in describing data variability, as evidenced by the sharp decline in variance that follows. The first few PCs are the emphasis of this understanding, which aids dimensionality reduction.
2. **PCA Variable Plot:** According to the PCA graph, PC1 and PC2 account for 76.7% of the variance in the data. Weight, Abdomen, and BodyFat are only a few of the variables that show strong correlations, with similar vector orientations signifying favourable relationships. Hip and thigh clustering indicates multicollinearity, which facilitates predictor selection.

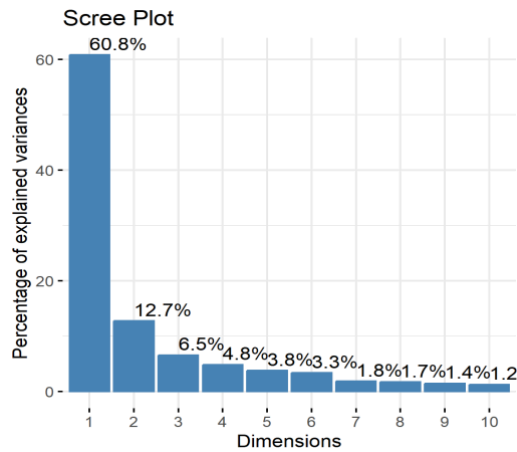


Figure 6: Scree plot

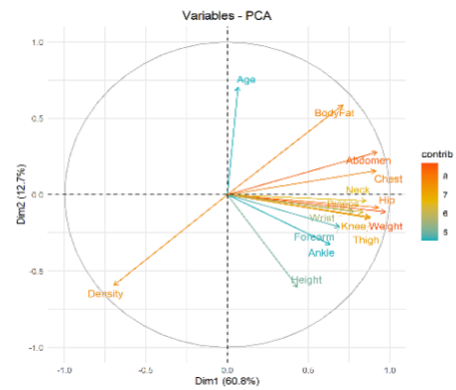


Figure 7: PCA Variable plot

ii. Hierarchical Clustering

Euclidean distance and Ward's approach are employed in Hierarchical Clustering to further investigate the clustering of observations according to body fat content. The clustering plot (Figure 8) displays the four unique clusters that were found in the results. Based on the major body measurements used in the analysis, these clusters show groups of people with comparable body fat compositions.

Interpretation: Based on the clustering results, body fat decomposition can be divided into four primary profiles, each of which may represent a distinct pattern of body composition or metabolism. This result is consistent with research showing how genetic and environmental factors impact body composition and obesity (Bouchard, 1994; Katzmarzyk & Mason, 2009).

Graphical Visualisation: The observations are arranged into four clusters, as shown in the Hierarchical Clustering Plot (Figure 8). This draws attention to the data's inherent groupings according to body fat characteristics.



Figure 8: Hierarchical Clustering Plot

4. Conclusion

Starting with data cleansing and correlation analysis, this study investigated body fat estimation using statistical analysis. The significance of anthropometric measurements was further supported by the

findings that the strongest predictors were density, abdomen, and chest. Multicollinearity was handled by regression models such as Lasso, and Kruskal-Wallis tests revealed notable age-related variations in body composition. Different body fat profiles were revealed using PCA and clustering, which also helped identify important factors affecting body fat distribution. These results provide information for individualised health evaluations and help to enhance predictive models.

5. References:

Borga, M., West, J., Bell, J. D., Harvey, N. C., Romu, T., & Heymsfield, S. B. (2018) 'Advanced body composition assessment: from body mass index to body composition profiling', *Journal of Investigative Medicine*, 66(5), pp. 1-10. Available at: <https://doi.org/10.1136/jim-2018-000722>.

Bouchard, C. (1994) *The Genetics of Obesity*. Boca Raton: CRC Press. Available at: <https://doi.org/10.1201/9781003068242>.

Bouchard, C., & Shepard, R. J. (1994) *Physical activity, fitness, and health: The model and key concepts*. Human Kinetics.

Chechik, G., Heitz, G., Elidan, G., Abbeel, P., & Koller, D. (2008) 'Max-margin classification of data with absent features', *The Journal of Machine Learning Research*, 9, pp. 1-21. Available at: <https://www.jmlr.org/papers/volume9/chechik08a/chechik08a.pdf>.

Derrick, B., Russ, B., Toher, D., & White, P. (2017) 'Test statistics for the comparison of means for two samples that include both paired and independent observations', *Journal of Modern Applied Statistical Methods*, 16(1), pp. 137-157. Available at: <https://doi.org/10.22237/jmasm/1493597280>.

Deurenberg, P., Weststrate, J. A., & Seidell, J. C. (1991) 'Body mass index as a measure of body fatness: age- and sex-specific prediction formulas', *British Journal of Nutrition*, 65(2), pp. 105-114. Available at: <https://doi.org/10.1079/BJN19910073>.

Freedman, D. S., Horlick, M., & Berenson, G. S. (2005) 'Body composition and adiposity indices in children and adolescents', *American Journal of Clinical Nutrition*, 82(6), pp. 1179-1186. Available at: https://journals.lww.com/coclinicalnutrition/Abstract/2005/11000/Body_mass_index_and_body_fatness_in_childhood.7.aspx#:~:text=This%20review%20covers%20the%20measurement%20and%20classification%20of,the%20relation%20of%20body%20mass%20index%20to%20adiposity.

Gallagher, D., Heymsfield, S. B., Heo, M., Jebb, S. A., Murgatroyd, P. R., & Sakamoto, Y. (2000) 'Healthy percentage body fat ranges: an approach for developing guidelines based on body mass index', *American Journal of Clinical Nutrition*, 72(3), pp. 694-701. Available at: <https://doi.org/10.1093/ajcn/72.3.694>.

Hawkins, D. M. (1980) *Identification of Outliers*. London: Chapman and Hall. Available at: <https://doi.org/10.1007/978-94-015-3994-4>.

Hsu, C. H., Liu, C. H., & Chang, T. H. (2003) 'Polynomial regression model applied to analyze the relationships among obesity indexes', *Obesity Research*, 11(4), pp. 485-491. Available at: <https://doi.org/10.1038/oby.2003.66>.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013) *An Introduction to Statistical Learning: With Applications in R*. New York: Springer. Available at: <https://doi.org/10.1007/978-1-4614-7138-7>.

Jolliffe, I. T. (2002) *Principal Component Analysis*. Springer. Available at: <https://link.springer.com/book/10.1007/b98835>.

- Jolliffe, I. T., & Cadima, J. (2016) 'Principal component analysis: a review and recent developments', *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 374(2065), p. 20150202. Available at: <https://doi.org/10.1098/rsta.2015.0202>.
- Kim, H.Y. (2013) 'Statistical notes for clinical researchers: assessing normal distribution using skewness and kurtosis', *Restorative Dentistry & Endodontics*, 38(1), pp. 52–54. Available at: <https://doi.org/10.5395/rde.2013.38.1.52>.
- Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2004) *Applied Linear Statistical Models* (5th ed.). McGraw-Hill. Available at: Applied linear statistical models: Kutner, Michael H., author: Free Download, Borrow, and Streaming: Internet Archive.
- Li, T., Hutfless, S., Scharfstein, D. O., Daniels, M. J., & Hogan, J. W. (2014) 'Standards should be applied in the prevention and handling of missing data for patient-centered outcomes research: a systematic review and expert consensus', *Journal of Clinical Epidemiology*, 67(1), pp. 15-32. Available at: <https://doi.org/10.1016/j.jclinepi.2013.08.013>.
- Little, R. J. A., & Rubin, D. B. (2019) *Statistical Analysis with Missing Data*. 3rd edn. Hoboken, NJ: Wiley. Available at: <https://doi.org/10.1002/9781119482260>.
- Samaras, T. (2007) 'Ageing and body fat: A review', *Ageing Research Reviews*, 6(1), pp. 76-91. Available at: <https://doi.org/10.1016/j.arr.2006.07.001>.
- Siri, W. E. (1961) 'Body composition from fluid spaces and density: analysis of methods', *Techniques for Measuring Body Composition*, 61, pp. 223-244. Available at: <https://pubmed.ncbi.nlm.nih.gov/8286893/>.
- Tibshirani, R. (1996) 'Regression shrinkage and selection via the Lasso', *Journal of the Royal Statistical Society: Series B (Methodological)*, 58(1), pp. 267-288. Available at: <https://doi.org/10.1111/j.25176161.1996.tb02080.x>.
- Tukey, J. W. (1977) *Exploratory Data Analysis*. Reading, MA: Addison-Wesley. Available at: <https://doi.org/10.2307/2683803>.
- Wilcox, R. R. (2011) *Introduction to Robust Estimation and Hypothesis Testing*. 3rd edn. Amsterdam: Academic Press. Available at: <https://www.perlego.com/book/1827405/introduction-to-robustestimation-and-hypothesis-testing-pdf>.